Stan Lee Natividad

AI Research Engineer

Mandaluyong City, Metro Manila 1441, PH · +63 991 153 2620 · leenatividad5@gmail.com · gryphlee.github.io · linkedin.com/in/stan-lee-natividad-83a327336

PROFILE

AI Research Engineer and 3rd-year undergraduate. Former confidential intelligence professional, with prior exposure to intelligence-related work, security analysis, and sensitive operational environments. Focused on advancing foundation models from passive language systems into capable reasoning and action-oriented agents. Researching continued pretraining, supervised fine-tuning (SFT), knowledge distillation, preference optimization, reinforcement learning, and agentic RL, with emphasis on training models through interaction with tools, simulators, and verifiable environments. Designing Agent–Environment Digital Twin architectures for state forking, counterfactual experimentation, simulation-based evaluation, and autonomous decision-making. Research interests span offensive and defensive cybersecurity, STEM and scientific reasoning, computational biology and laboratory AI, coding and robotics, agriculture, and critical infrastructure. Conducting applied AI research and leading research-paper development, from research question formulation and experimental design to implementation, evaluation, and analysis. Complemented by hands-on expertise in LLM evaluation, adversarial red-teaming, GPU infrastructure, distributed training, and production model serving.

EMPLOYMENT HISTORY

AI Research Engineer | Kooya Inc *Oct 2026 — Present*

BGC, Taguig

- Leading research and development of a Filipino-focused open foundation-model ecosystem, targeting multilingual, Taglish, STEM, coding, cybersecurity, and robotics capabilities.
- Designing the model-development pipeline across continued pretraining, supervised fine-tuning, synthetic data generation, preference optimization, and reinforcement learning.
- Building large-scale Filipino and regional-language datasets for Tagalog, Taglish, Cebuano, Ilocano, Kapampangan, and other Philippine languages.
- Designing evaluation frameworks for language capability, reasoning, coding, cybersecurity, and domain-specific performance.
- Researching agentic reinforcement learning and digital-twin environments for counterfactual experimentation, state forking, and autonomous decision-making.
- Exploring specialized AI systems for robotics, cybersecurity, coding, disaster response, agriculture, and critical infrastructure.

AI Infrastructure Engineer & Machine Learning Engineer | Kooya Inc *Jun 2026 — Oct 2026*

BGC, Taguig

- Managed the AI platform used by engineering teams to fine-tune and serve models — GPU compute allocation, distributed training with Ray, and high-performance inference via vLLM.
- Fine-tuned open-source LLMs (LoRA, QLoRA, SFT, knowledge distillation, GRPO, DPO) end-to-end, from dataset preparation through evaluation and production-readiness review.
- Maintained and extended the company's production RAG pipeline and multi-agent orchestration, serving 4 internal teams and ~1,200 queries per day.
- Built observability across serving infrastructure, RAG pipelines, and local models using RAGAS and DeepEval — cutting measured hallucination rate from 12% to 4% and p95 latency from 2.1s to 850ms.
- Red-teamed in-house models before deployment (jailbreak and prompt/system-injection testing), surfacing and patching 15+ exploitable behaviors pre-launch.
- Built a reproducible benchmarking environment comparing open-source models on reasoning quality, latency, and cost — informing model selection for the company's internal knowledge assistant.
- Supported the migration from third-party APIs to locally hosted models, projected to cut inference costs by ~60% (≈₱180K/month).

Intelligent Systems Engineer | Power4All (6-month project engagement) *Jan 2026 — Jun 2026*

Pasig

- Designed and delivered a fully licensed, company-only local AI model for the engineering and business development teams, replacing third-party AI services for document drafting, technical Q&A, and proposal support.

- Built a centralized company knowledge base with an end-to-end RAG pipeline, unifying company files and operational data into a single retrieval backbone.
- Automated the bid-discovery workflow for business development — auto-scraping opportunities from external sites into an internal platform, saving an estimated 10 hours per week of manual searching.
- Built a digital twin of the company's water-treatment plant, enabling the team to simulate chemical dosing in software before applying changes to the physical plant — reducing trial-and-error dosing adjustments by ~80%.
- Initiated broader Industry 4.0 work including AI-integrated SCADA monitoring and workflow automation across supply chain and engineering functions.

Enterprise AI Architect | Green Attic Roofing *Dec 2024 — Dec 2025*

Chicago (Remote)

- Converted manual, repetitive operations into automated AI-driven workflows across 3 departments, saving an estimated 25+ hours per week of staff time.
- Built internal AI tools on request from teams across the business — integrating LLMs, external APIs, and internal databases into unified workflows.
- Created reusable AI frameworks that cut build time for new automations from ~2 weeks to ~3 days.
- Served as the organization's AI adoption lead — evaluating tools and delivering solutions built for security and maintainability.

AI Engineer | Silicon Valley Associate *Jan 2024 — Nov 2024*

Remote (UAE-based consultancy)

- Built an AI-powered hiring platform with LLM-driven candidate matching, screening, and routing, processing 2,000+ applications.
- Designed and shipped an AI interviewer backed by a full RAG pipeline — retrieving role-specific context to drive dynamic questioning and evaluation, reducing time-to-screen by ~70%.
- Worked directly with the CEO to convert manual hiring workflows into scalable agentic systems.

EDUCATION

BS Electronics Engineering | Rizal Technological University *Mandaluyong*

3rd-year student (ongoing)

CORE SKILLS

Foundation Models & Post-Training: Continued Pretraining · SFT · LoRA / QLoRA / Adapters · Knowledge Distillation · DPO · GRPO · RLHF · Model Quantization · Synthetic Data Generation · Model Adaptation & Domain Specialization

Agentic AI & Reinforcement Learning: Agentic AI · Reinforcement Learning · Agentic RL · Tool Use & Function Calling · Multi-Agent Systems · Agent Orchestration · Environment-Grounded Learning · Verifiable Reward & Feedback Systems · Simulation-Based Training

Digital Twins & AI Environments: Agent–Environment Digital Twins · State Forking · Counterfactual Experimentation · Digital Twin Simulation · Simulation-Based Evaluation · Autonomous Decision Environments · AI-SCADA

AI Research & Evaluation: Experimental Design · Model Benchmarking · Ablation Studies · Error Analysis · RAG Evaluation · RAGAS · DeepEval · Groundedness & Faithfulness · Hallucination Analysis · Robustness Testing · Research Data Curation

AI Security: Defensive & Controlled Offensive Security Research · Adversarial Testing & Red-Teaming · Jailbreak & Prompt-Injection Testing · Model Security · Secure AI Systems · Reverse Engineering & Binary Analysis

Domain Research: STEM & Scientific Reasoning · Computational Biology · Laboratory AI · Coding & Robotics · Embedded Systems · Agriculture AI · Critical Infrastructure AI · Disaster & Resilience Systems

RAG & Knowledge Systems: End-to-End RAG Pipelines · Retrieval & Reranking · Knowledge Bases · Hybrid Search · Multi-Modal / Document Intelligence · Production RAG Systems

AI Infrastructure & Model Serving: GPU Computing · Distributed Training (Ray) · PyTorch · Hugging Face · vLLM · ONNX / TensorRT · Local & Self-Hosted Model Deployment · Kubernetes (working knowledge)

Automation & Industrial AI: Industry 4.0 · Workflow Automation · n8n · Zapier · AI-Driven Industrial Systems · Digital Twin Integration